

1. Anchor Bolts segment

input

- N_b = total number of anchor bolts
- db = nominal anchor bolt diameter (in)
- dh = anchor bolt hole diameter (in)
- rb = anchor bolt circle radius (ft or in)
- α_b = start angle of first anchor bolt (deg)
- θ_{seg} = segment angle (deg), if bolts are checked by segment
- n_s = number of segments
- t_{bp} = base plate thickness (in)
- f_y = bolt yield strength
- f_u = bolt ultimate strength
- T_u = design uplift/tension demand on one bolt or bolt group
- V_u = design shear demand on one bolt or bolt group
- ϕ = resistance factor, if your software includes strength check
- e = edge distance from hole to plate edge
- s = bolt spacing, if user enters directly
- nbs = number of bolts per segment

Output

For B-Change-100K G:

Given

Tank radius: $R = 10'6'' \text{ ft}$

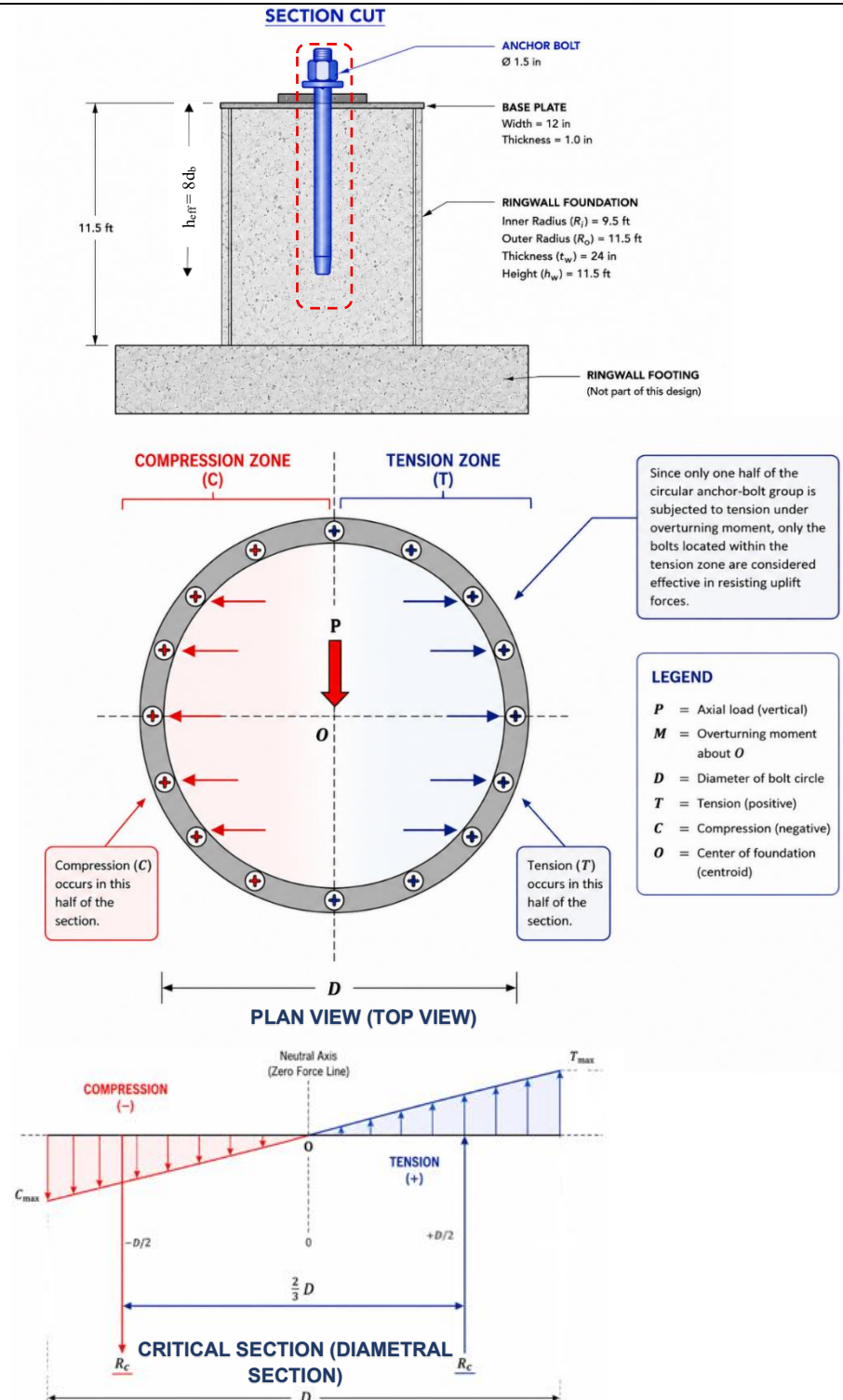
Governing moment: $M_u = 4847.67 \text{ kip}$

–ftft,

Inner face: $10.5 - 1 = 9.5 \text{ ft}$ (matches 9'–6")

Outer face: $10.5 + 1 = 11.5 \text{ ft}$

Ring wall thickness = 2 ft



1. Governing Overturning Moment: From calculation

$$M_u = 4847.67 \text{ kip-ft}$$

Diameter of bolt circle:

$$D = 21 \text{ ft}$$

Effective lever arm:

$$0.67D = 0.67(21) = 14.07 \text{ ft}$$

Total uplift force:

$$T_u = \frac{M_u}{0.67D}$$

$$T_u = \frac{4847.67}{14.07}$$

$$T_u = 344.54 \text{ kips}$$

2. Tension Force Per Bolt

You used:

$$T_{bolt} = \frac{2T}{N_b}$$

This is correct for a simplified circular anchor group assumption.

Given:

$$N_b = 20$$

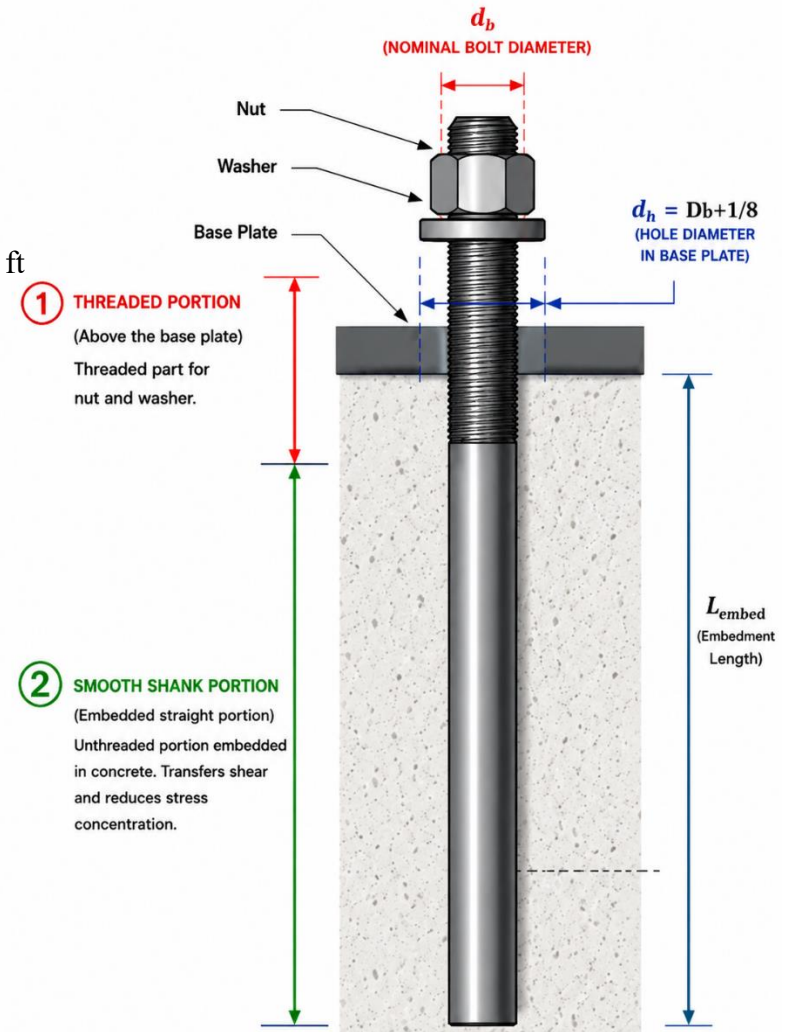
$$T_{bolt} = \frac{2(344.54)}{20}$$

$$T_{bolt} = 34.45 \text{ kips}$$

3. Bolt Area

We choose reasonable trial values:

- $N_b = 20$ bolts
- $d_b = 1.5 \text{ in.}$ (In accordance with ACI 318 anchor design provisions the anchor-bolt diameter d_b is taken as the nominal bolt diameter, corresponding to the major diameter of the threaded anchor rod.)

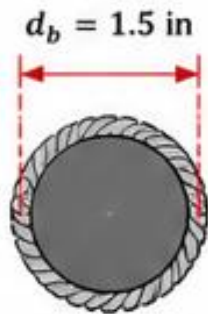


- $d_h = 1.625 \text{ in } (1/8 + d_b)$
- $r_b = 10.75 \text{ ft} = 129 \text{ in}$
- $\alpha_b = 0^\circ$

$$A_b = \frac{\pi d_b^2}{4}$$

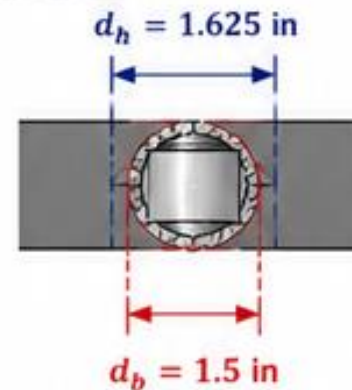
$$A_b = \frac{\pi (1.5)^2}{4}$$

$$A_b = 1.767 \text{ in}^2$$



HOW $d_b = 1.5 \text{ in}$ IS CONSIDERED

d_b is the nominal bolt diameter (measured at the major diameter of the threaded portion). It is the standard size of the anchor bolt used for design calculations (tension, shear, bearing, etc.).



d_h is the diameter of the hole in the base plate. It is kept slightly larger than d_b to allow for installation tolerance and easy erection.

4. Hole Area

$$A_h = \frac{\pi d_h^2}{4}$$

$$A_h = \frac{\pi (1.625)^2}{4}$$

$$A_h = 2.074 \text{ in}^2$$

5. Angular Spacing

$$\Delta\theta_b = \frac{360}{N_b}$$

$$\frac{360}{20} = 18^\circ$$

6. Bolt Coordinates

For bolt i :

$$\theta_i = (i - 1) \times 18^\circ$$

$$x_i = r_b \cos \theta_i$$

$$y_i = r_b \sin \theta_i$$

e.g. if $i=1$ $x_i=0$, Where: $r_b = 10.75 \text{ ft}$

If $i=2$

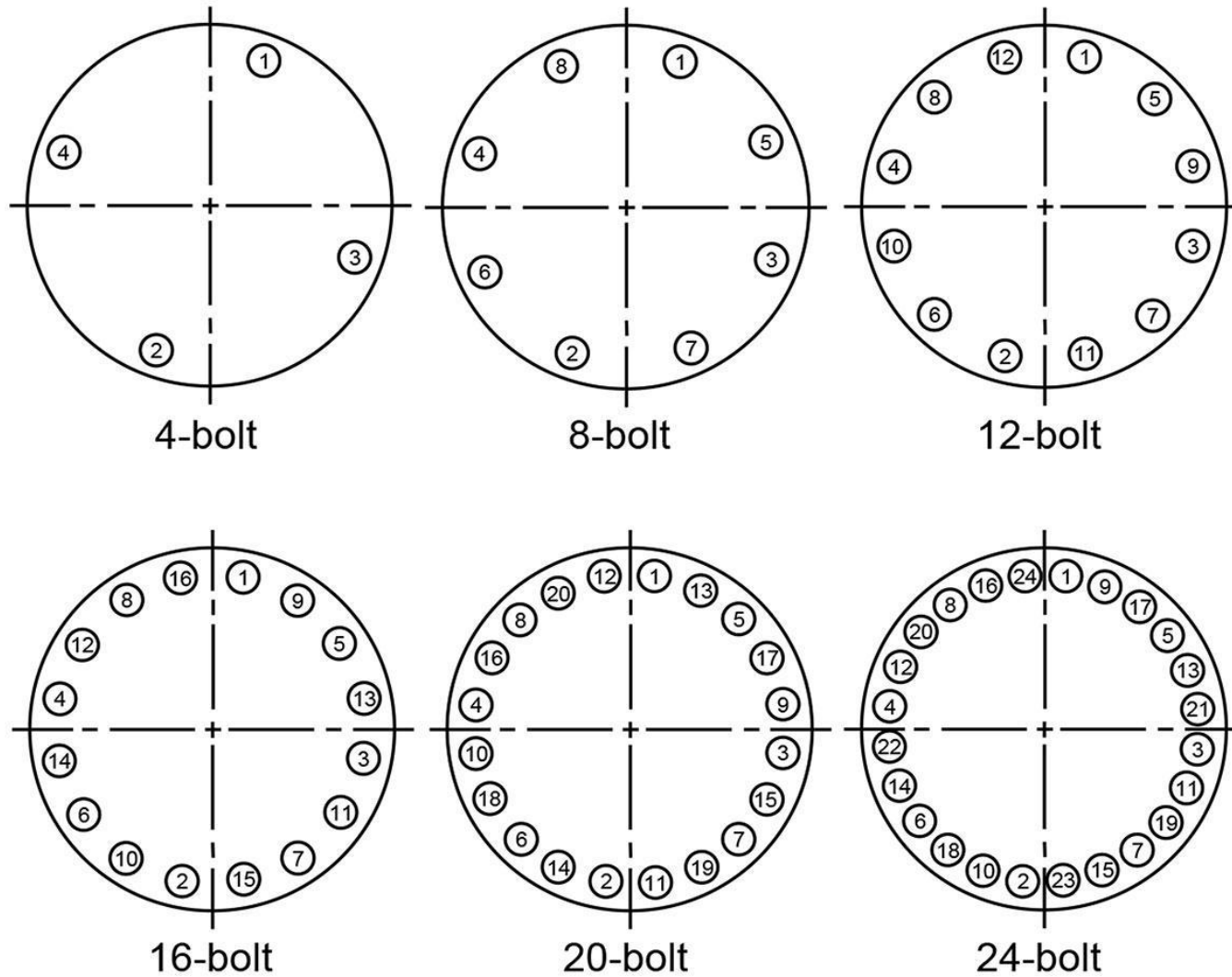
$$\theta_i = (2 - 1) \times 18^\circ = 18^\circ$$

$$x_2 = 10.75 \text{ ft} \times \cos 18^\circ = 10.23 \text{ ft}$$

$$y_2 = 10.75 \text{ ft} \times \sin 18^\circ = 3.32 \text{ ft}$$

Anchor Bolt Layout Table

Bolt	Angle (°)	x (ft)	y (ft)	Location
1	0	10.75	0	Right
2	18	10.23	3.32	Upper-right
3	36	8.7	6.32	Upper-right
4	54	6.32	8.7	Upper-right
5	72	3.32	10.23	Upper-right
6	90	0	10.75	Top
7	108	-3.32	10.23	Upper left
8	126	-6.32	8.7	Upper left
9	144	-8.7	6.32	Upper left
10	162	-10.23	3.32	Upper left
11	180	-10.75	0	Left
12	198	-10.23	-3.32	Lower left
13	216	-8.7	-6.32	Lower left
14	234	-6.32	-8.7	Lower left
15	252	-3.32	-10.23	Lower left
16	270	0	-10.75	Bottom
17	288	3.32	-10.23	Lower-right
18	306	6.32	-8.7	Lower-right
19	324	8.7	-6.32	Lower-right
20	342	10.23	-3.32	Lower-right



7. Arc Spacing

Use radius in inches:

$$r_b = 129 \text{ in}$$

$$S_{arc} = r_b \Delta\theta_b \frac{\pi}{180}$$

$$S_{arc} = 129(18) \frac{\pi}{180}$$

$$S_{arc} = 40.53 \text{ in}$$

8. Chord Spacing

$$S_{chord} = 2r_b \sin \left(\frac{\Delta\theta_b}{2} \right)$$

$$S_{chord} = 2(129) \sin (9^\circ)$$

$$S_{chord} = 40.35 \text{ in}$$

9. Number of Bolts Per Segment

$$n_{bs} = \frac{N_b}{n_s}$$

Typical choices:

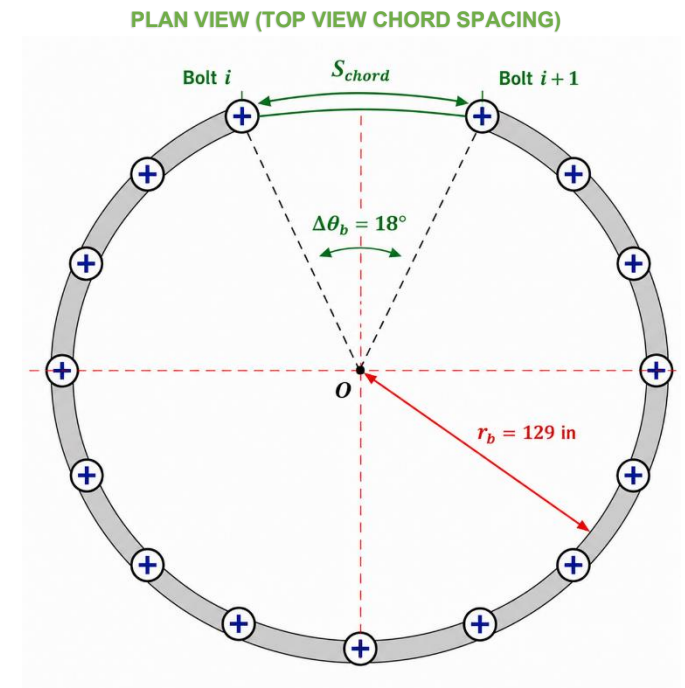
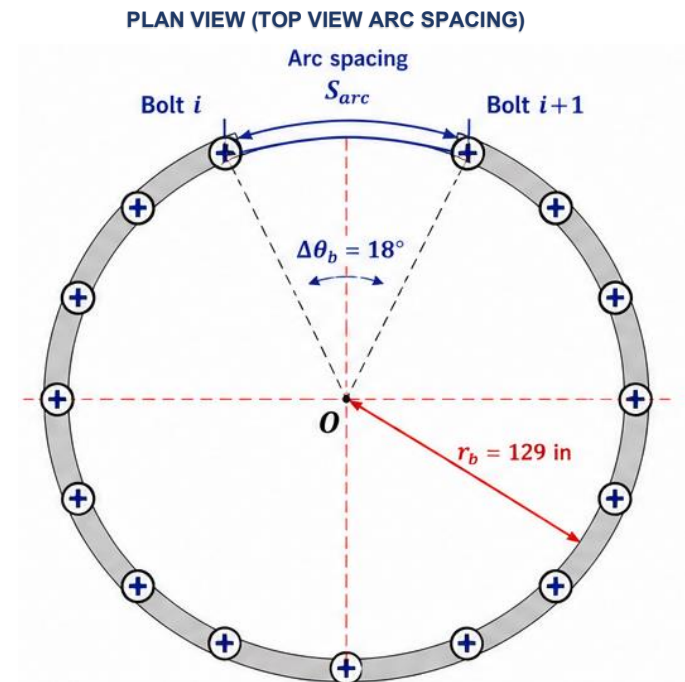
- 4 segments (quadrants)
- 6 segments (60° each)
- 8 segments (45° each)

Case 1: 6 Segments

If:

- $N_b = 20$
- $n_s = 6$

$n_{bs} = 3.33 \dots$ Not Acceptable (Software Shouldn't accept this or Flag it).



Case 2: 4 Segments

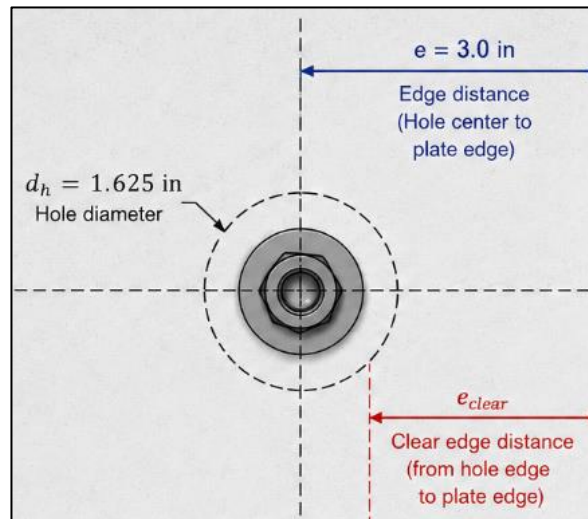
- $n_{bs} = \frac{20}{4} = 5 \dots \dots \dots \text{Acceptable}$

Case 3: 5 Segments

- $n_{bs} = \frac{20}{5} = 4 \dots \dots \dots \text{Acceptable}$

10. Bolt Clear Edge Distance Check

If the hole is located between plate edges, the minimum clear edge distance can be written as:



$$d_h = 1.625 \text{ in}$$

$$e_{clear} = e - \frac{d_h}{2}$$

$$e_{clear} = 3.0 - \frac{1.625}{2}$$

$$e_{clear} = 3.0 - 0.8125$$

$$e_{clear} = 2.1875 \text{ in}$$

$$e_{clear} = \mathbf{2.19 \text{ in}}$$

Check Against Code

From ACI / AISC / practical detailing:

Typical minimums:

- $\geq 1.5 \text{ in}$ (absolute minimum)
- $\geq 1.5 \times \text{bolt diameter}$ (preferred)

For your bolt:

$$1.5 \times d_b = 1.5 \times 1.5 = 2.25 \text{ in}$$

$$e_{clear} = 2.19 \text{ in} > 1.5 \text{ in} \dots \dots \dots \checkmark \text{SAFE}$$

Recommend: Increase edge distance to 3.25–3.5 in

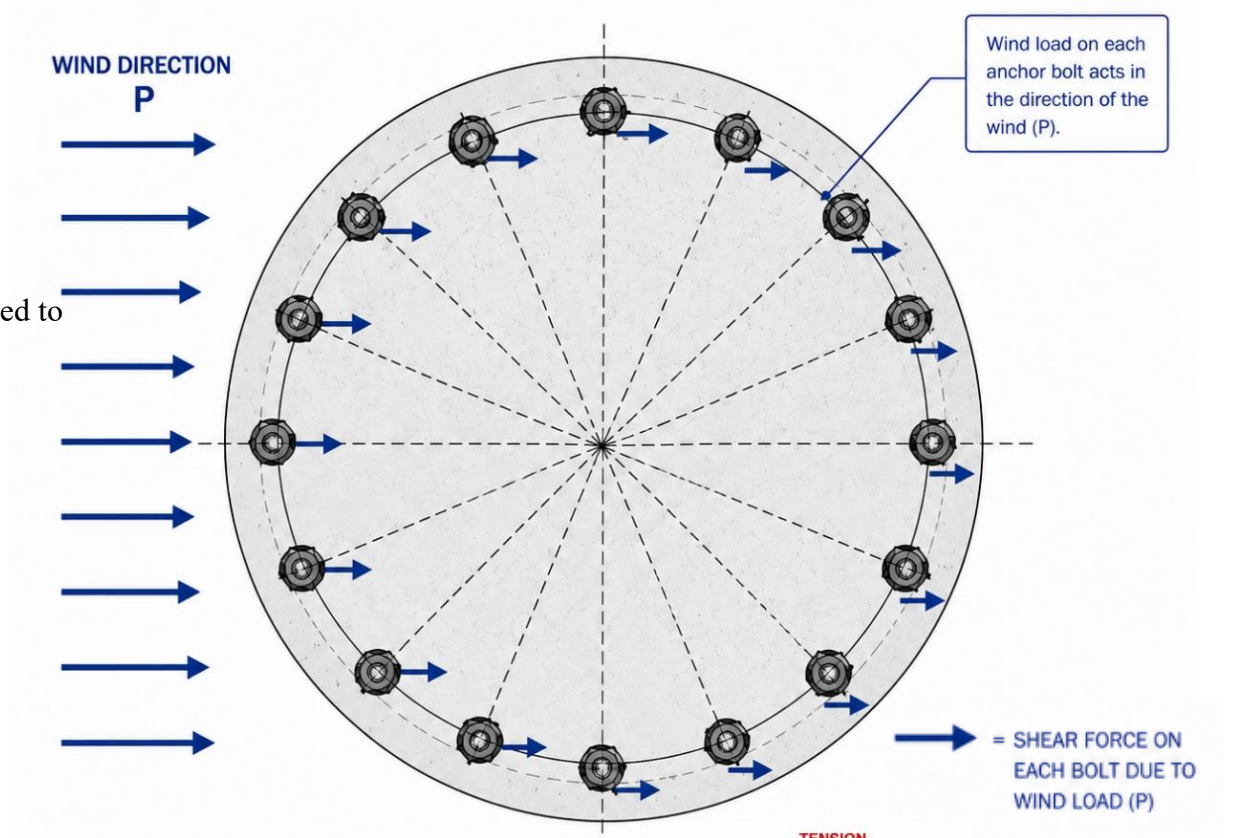
11. Shear Per Bolt

From your data: Wind shear ≈ 20.683 kips

Each Bolt Resists Shear: All anchor bolts are subjected to shear force parallel to the wind direction (P).

Per bolt:

$$V_{bolt} = \frac{20.683}{20}$$
$$V_{bolt} = 1.034 \text{ kips}$$



12. Tensile Capacity

Using:

$$\phi P_{nt} = \phi A_b F_u$$

Given:

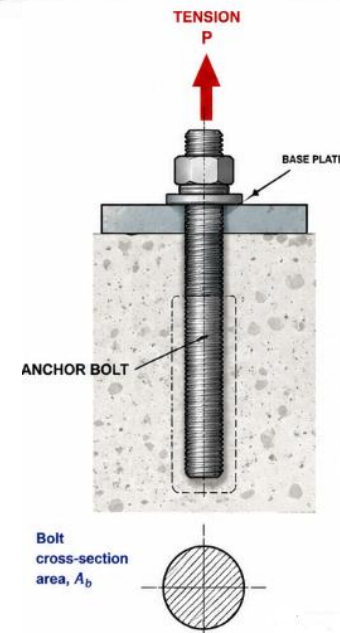
$$\phi = 0.75$$

$$A_b = 1.767$$

$$F_u = 58 \text{ ksi}$$

$$\phi P_{nt} = 0.75(1.767)(58)$$

$$\phi P_{nt} = 76.87 \text{ kips}$$



13. Shear Capacity

$$F_{nv} = 0.6F_u$$

$$F_{nv} = 0.6(58) = 34.8 \text{ ksi}$$

$$\phi V_n = 0.75(1.767)(34.8)$$

$$\phi V_n = 46.11 \text{ kips}$$

14. Interaction Check

$$\text{Utilization} = \left(\frac{N_u}{\phi N_n}\right)^\alpha + \frac{V_u}{\phi V_n} \leq 1 \dots \dots \dots \text{ACI 318-19 Sec 17.6}$$

$$\text{Utilization} = \frac{T_{bolt}}{\phi P_{nt}} + \frac{V_{bolt}}{\phi V_n} \leq 1$$

$$= \frac{34.45}{76.87} + \frac{1.034}{46.11}$$

$$= 0.448 + 0.022$$

$$= 0.470$$

$$\text{Utilization} = 0.47 < 1.0 \Rightarrow \text{SAFE}$$

15. Anchor Bolt Length Design:

- Minimum embedment (code-based)

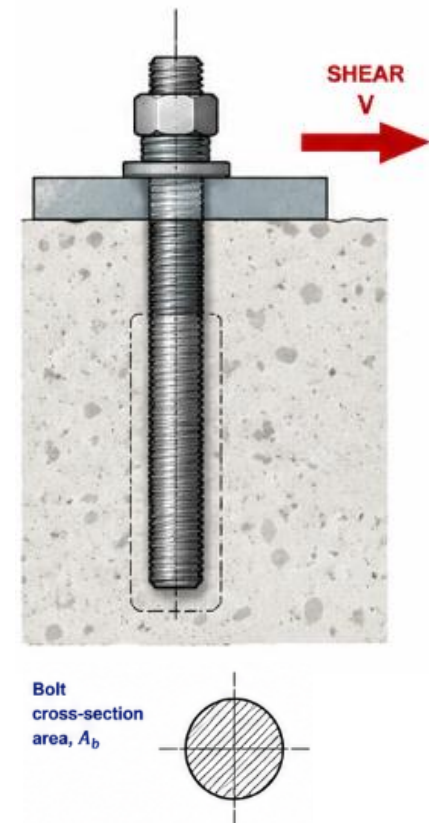
ACI / AISC / industry:

$$h_{ef} \geq 8d_b$$

$$h_{ef} = 8 \times 1.5 = 12 \text{ in}$$

16. Pull Out Test/ Concrete Breakout Capacity

$$N_b = k_c \sqrt{f'_c} h_{ef}^{1.5}$$



where:

- N_b = concrete breakout strength
- k_c = empirical constant
- f'_c = concrete compressive strength
- h_{ef} = effective embedment depth.
- Embedment = approximately 9–10 ft
- Effective embedment = assume $h_{ef} = 108$ in conservative

Based on the foundation detail drawing,

$$h_{ef} = 96 \text{ in}$$

Substituting:

$$N_b = 24\sqrt{4000}(96)^{1.5} = 24(63.25)(940.6)$$

$$N_b = 1,427,470 \text{ lb}$$

$$N_b \approx 1427 \text{ kips}$$

Using the ACI strength reduction factor:

$$\phi = 0.75$$

$$\phi N_b = 0.75(1427)$$

$$\phi N_b \approx 1070 \text{ kips}$$

Maximum tensile demand per anchor bolt:

$$T_{bolt} = 34.45 \text{ kips}$$

17. Concrete breakout utilization:

$$Utilization = \frac{T_{bolt}}{\phi N_b}$$

$$= \frac{34.45}{1070}$$

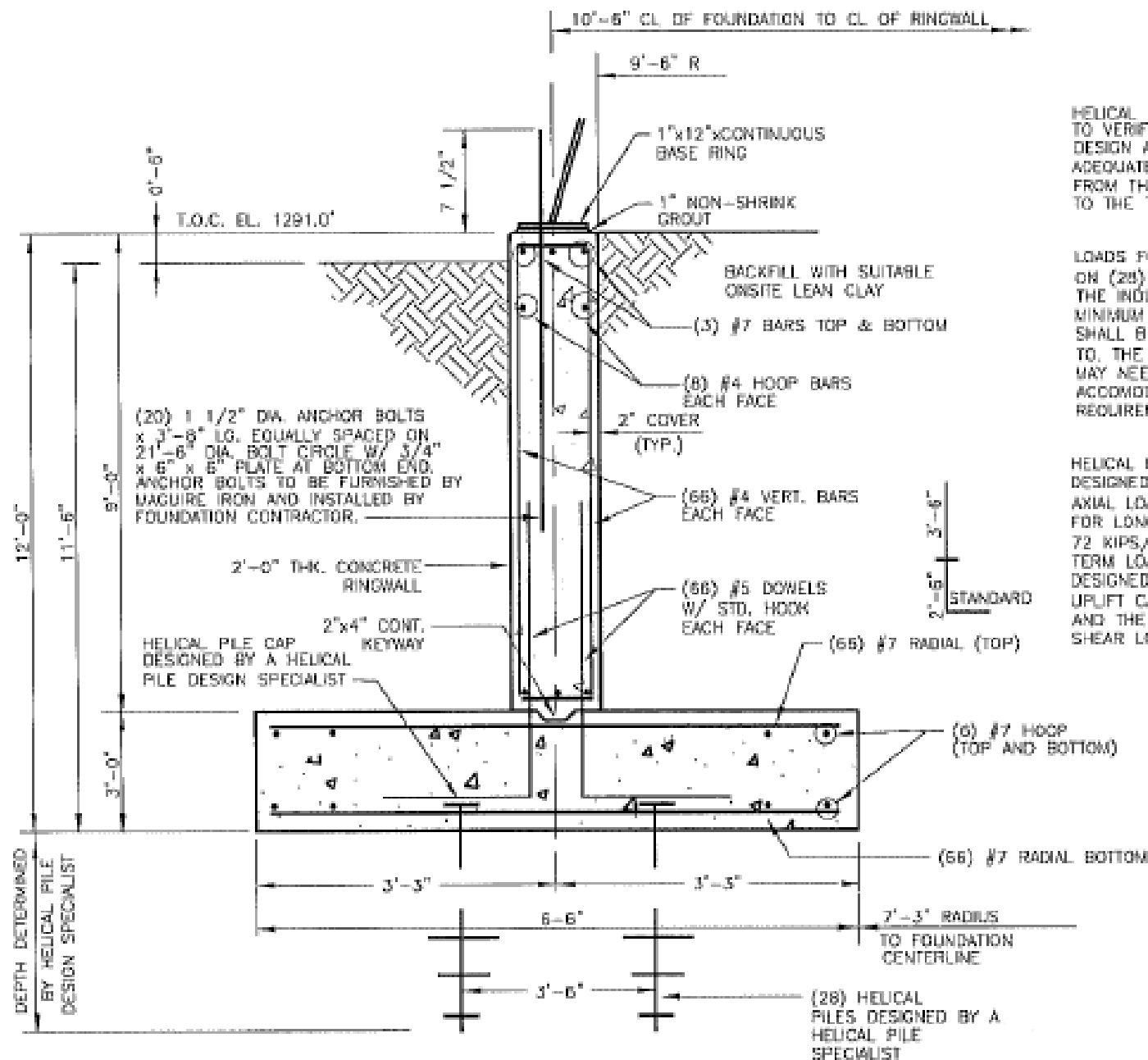
$$= 0.032$$

$$0.032 < 1.0$$

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Parameter	Value
Number of anchor bolts, (N_b)	20
Anchor bolt diameter, (d_b)	1.5 in
Hole diameter, (d_h)	1.625 in
Bolt circle radius, (r_b)	10.75 ft (129 in)
Angular spacing, ($\Delta\theta_b$)	18°
Arc spacing, (S_{arc})	40.53 in
Chord spacing, (S_{chord})	40.35 in
Clear edge distance, (e_{clear})	2.19 in
Bolt tensile area, (A_b)	1.767 in ²
Bolt hole area, (A_h)	2.074 in ²
Tension demand per bolt, (T_{bolt})	34.45 kips
Shear demand per bolt, (V_{bolt})	1.03 kips
Tensile capacity, ϕP_{nt}	76.87 kips
Shear capacity, ϕV_n	46.11 kips
Interaction utilization ratio	0.47
Governing interaction equation	Utilization = $\frac{T_{bolt}}{\phi P_{nt}} + \frac{V_{bolt}}{\phi V_n} \leq 1$
Governing design check	Combined tension and shear
Concrete breakout check	Not governing
Anchor embedment behavior	Deep cast-in plate-anchor system
Final anchor bolt design status	✓ SAFE

Note: The present calculations primarily evaluate steel anchorage strength and preliminary anchorage behavior. Final design should also verify reinforcement detailing, pedestal confinement, local concrete stresses, and constructability requirements in accordance with ACI 318 Chapter 17 and project-specific structural drawings.



HELICAL PILE SPECIALIST TO VERIFY THAT THE RINGWALL DESIGN AND REINFORCEMENT IS ADEQUATE TO TRANSFER ALL LOADS FROM THE HELICAL PILES TO THE TOWER.

LOADS FOR PILES ARE BASED ON (28) HELICAL PILES. THE INDUSTRY STANDARD MINIMUM SPACING REQUIREMENTS SHALL BE STRICTLY ADHERED TO. THE QUANTITY AND LOADS/PILE MAY NEED TO BE ADJUSTED TO ACCOMMODATE THAT SPACING REQUIREMENT.

HELICAL PILES TO BE DESIGNED FOR AN ALLOWABLE AXIAL LOAD OF 72 KIPS/PILE, FOR LONG TERM LOADS AND 72 KIPS/PILE FOR SHORT TERM LOADS. THE PILE TO BE DESIGNED ALSO FOR AN ALLOWABLE UPLIFT CAPACITY OF 15 KIPS/PILE AND THE SYSTEM FOR A TOTAL SHEAR LOAD OF 30 KIPS.